For Regular Graphs, Galvin's Kernel Method Works Only on Bipartite Graphs

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Abstract

Galvin proved the list edge colouring conjecture for bipartite graphs by exhibiting, for every bipartite graph G of maximum degree Δ, an orientation of the line graph $L(G)$ which is kernel-perfect and has every out-degree at most $\Delta - 1$. The kernel lemma of Bondy, Boppana and Siegel turns such an orientation into a list edge colouring from lists of size Δ. We prove the reverse direction for regular graphs: if G is a simple k-regular graph and $L(G)$ has a kernel-perfect orientation with every out-degree at most $k - 1$, then G is bipartite. Together with Galvin's theorem this gives a characterisation: for simple regular graphs, the kernel method with lists of size Δ succeeds exactly on the bipartite graphs. We also prove that on a regular bipartite graph every such orientation is one of Galvin's own orientations, so for regular graphs the kernel method contains nothing beyond Galvin's construction. Consequently the kernel method cannot prove the list edge colouring conjecture for any non-bipartite regular graph of class one, a family that contains the even complete graphs and the 52 regular graphs on at most 10 vertices that Schauz left open and that no known theorem covers. The proof is elementary. It is written here in very small steps, with every lemma proved from the definitions, so that it can be checked line by line. Every number quoted is reproduced by a script in the repository.

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1 How to read this document

This document exists so that the proof can be checked. It is written for a reader who wants to verify every line, and it therefore proves many facts that a research article would call standard. Nothing in the proof of the main results is left to the reader.

What is proved and what is cited. The following are *cited without proof*. None of them is used in the proof of the main theorem (Theorem 2.6) or in the proof of the rigidity theorem (Theorem 16.3), except where explicitly marked.

(C1) The kernel lemma (Lemma 2.3). It explains why the main theorem matters. It is not used in any proof.

(C2) Galvin's theorem (Theorem 2.5), the forward direction. It is used in exactly one place, the surjectivity half of Theorem 16.3, and this is marked there.

(C3) Results of other authors quoted in Section 18 when discussing consequences. These concern the significance of the theorem and not its truth.

Everything else, including every lemma about walks, cycles, bipartite graphs, digraphs, kernels and tournaments that the proof uses, is proved in Section 3.

How the proof is laid out. The proof of Theorem 2.6 has nine steps. Each step is one section from Section 5 to Section 13, and each section ends with a single displayed conclusion that the later sections use. Table 1 lists the steps. A verifier can check one step at a time, taking the conclusions of earlier steps as given. The dependency of every lemma on earlier lemmas is listed in Section A.

Conventions. All graphs are finite. We write $a \equiv b$ for $a \equiv b \pmod 2$, unless another modulus is written. "Iff" means "if and only if". When indices run around a cycle of length m, they are read modulo m, so that $a_{m+1} = a_1$ and $a_0 = a_m$.

2 The problem and the results

2.1 List edge colouring

Let G be a graph. An *edge colouring* of G gives each edge a colour so that two edges with a common end receive different colours. The least number of colours for which this is possible is the *chromatic index* $\chi'(G)$. Since the edges at a vertex of degree d need d different colours, $\chi'(G) \geq \Delta(G)$, where $\Delta(G)$ is the maximum degree. Vizing [12] proved $\chi'(G) \leq \Delta(G) + 1$ for every simple graph. A simple graph is of *class one* if $\chi'(G) = \Delta(G)$ and of *class two* if $\chi'(G) = \Delta(G) + 1$.

In *list* edge colouring each edge e is given its own list $C(e)$ of allowed colours, and we ask for an edge colouring in which each edge takes a colour from its own list. The *list chromatic index* $\mathrm{ch}'(G)$ is the least k such that this is possible whenever every list has at least k colours. Giving every edge the same list $\{1, \dots, k\}$ is ordinary edge colouring, so $\mathrm{ch}'(G) \geq \chi'(G)$ always.

Conjecture 2.1 (List Edge Colouring Conjecture)**.** *For every graph G,* $\mathrm{ch}'(G) = \chi'(G)$.

The *line graph* $L(G)$ has the edges of G as its vertices, two of them adjacent when they share an end. Edge colourings of G are exactly vertex colourings of $L(G)$, so the conjecture says that list colouring is no harder than ordinary colouring on line graphs.

2.2 Kernels and the kernel lemma

Definition 2.2 (kernel)**.** Let D be a digraph. A set K of vertices of D is a *kernel* if

(K1) no arc of D joins two vertices of K, and

(K2) every vertex x not in K has an arc $x \to z$ with $z \in K$.

D is *kernel-perfect* if for every set S of vertices, the subdigraph induced on S has a kernel.

The following lemma is the engine of the kernel method. It is due to Bondy, Boppana and Siegel (unpublished), and appears with proof in Alon and Tarsi [2] and in the chapter on the Dinitz problem in Aigner and Ziegler [1]. Here $d^+_D(x)$ is the number of arcs leaving x.

Lemma 2.3 (kernel lemma; cited, not used in any proof)**.** *Let H be a graph and let D be a kernel-perfect orientation of H. If every vertex x of H has a list $C(x)$ with $|C(x)| \geq d^+_D(x) + 1$, then H can be properly coloured with each vertex taking a colour from its own list.*

Applied to $H = L(G)$, the lemma says: *if $L(G)$ has a kernel-perfect orientation in which every out-degree is at most $k - 1$, then* $\mathrm{ch}'(G) \leq k$. We name the object it asks for.

Definition 2.4 (k-certificate)**.** Let G be a graph and $k \geq 1$ an integer. A *k-certificate* for G is an orientation D of $L(G)$ which is kernel-perfect and in which $d_D^+(e) \leq k-1$ for every edge e of G.

So a k-certificate proves $\mathrm{ch}'(G) \leq k$. The kernel method, in the form used by Galvin, is the search for a $\chi'(G)$-certificate: if one exists, then $\chi'(G) \leq \mathrm{ch}'(G) \leq \chi'(G)$ and the conjecture holds for G.

Theorem 2.5 (Galvin [5]; the forward direction, cited)**.** *If G is a bipartite graph of maximum degree Δ, then G has a Δ-certificate. Consequently* $\mathrm{ch}'(G) = \chi'(G) = \Delta$.

Galvin's theorem holds for bipartite multigraphs as well. The proof in [5] and in [1] constructs the certificate explicitly. We recall the construction in Definition 16.1, because the rigidity theorem says that, for regular graphs, it is the only one.

2.3 The main results

Theorem 2.6 (main theorem)**.** *Let $k \geq 1$ and let G be a simple k-regular graph. If G has a k-certificate, then G is bipartite.*

Combining Theorem 2.6 with Theorem 2.5 (for a k-regular graph, $\Delta = k$) gives:

Corollary 2.7 (characterisation)**.** *Let $k \geq 1$ and let G be a simple k-regular graph. Then $L(G)$ has a kernel-perfect orientation with every out-degree at most $k-1$ if and only if G is bipartite.*

Corollary 2.8 (the barrier)**.** *Let G be a simple regular graph which is not bipartite. Then the kernel lemma, applied to $L(G)$ with lists of size $\Delta(G)$, cannot prove* $\mathrm{ch}'(G) = \Delta(G)$*, whatever the true value of* $\mathrm{ch}'(G)$ *is.*

Proof. To apply the kernel lemma with lists of size $\Delta(G)$ one needs a $\Delta(G)$-certificate. By Theorem 2.6 none exists. □

The second main result says that on regular bipartite graphs the certificates are exactly Galvin's. It is stated precisely as Theorem 16.3 in Section 16, after Galvin's orientations are defined.

Theorem 2.9 (rigidity, informal)**.** *If G is simple, k-regular and bipartite, then every k-certificate for G is a Galvin orientation, built from a proper k-edge colouring in the way Galvin builds it. The certificates are in bijection with the proper edge colourings with colours $0, 1, \ldots, k-1$.*

Finally, the main theorem extends to the setting in which an edge of $L(G)$ may be oriented in both directions at once (Proposition 17.1).

2.4 What the main theorem does not say

Theorem 2.6 is about a *method*, not about the *value* of $\mathrm{ch}'(G)$. It does *not* say that $\mathrm{ch}'(G) > \Delta(G)$ for non-bipartite regular G. That statement is false: K_4 is 3-regular and not bipartite, yet $\mathrm{ch}'(K_4) = 3$ (this follows from the Alon and Tarsi method [2], and also from the theorem of Ellingham and Goddyn [4], since K_4 is planar). What Theorem 2.6 says about K_4 is that this equality cannot be proved by the kernel lemma with lists of size 3. Further limits of the statement are listed in Section 18.5.

2.5 Why the reverse direction is not automatic

A natural first reaction is: Galvin's method already needs G to be bipartite, so what does the reverse direction add? The answer is that bipartiteness is a hypothesis of Galvin's *construction*, not of the kernel *method*.

(i) The kernel lemma (Lemma 2.3) holds for every digraph. Nothing in it mentions bipartite graphs.

(ii) Galvin uses bipartiteness to *build* one particular orientation. He takes a proper Δ-edge colouring, which exists by König's theorem [7], and reads the colours upwards on one side of the bipartition and downwards on the other. That this recipe needs two sides says nothing about whether some other recipe could work on a graph without two sides.

(iii) Other recipes do work outside the bipartite world. McDonald [8, Theorem 3] proves that if G is a bipartite graph plus one edge, then $L(G)$ has a $\chi'(G)$-certificate. The smallest example is the diamond, K_4 minus an edge: it is not bipartite, it is class one with $\Delta = 3$, and a brute-force search finds exactly two 3-certificates for it (Table 2). So the kernel method *can* prove the conjecture for some non-bipartite graphs with lists of size Δ.

(iv) With one more colour, the kernel method reaches further still: McDonald proves that the Petersen graph (class two, so $\chi' = \Delta + 1 = 4$) has a 4-certificate [8, Theorem 4], and that every simple graph without odd cycles of length at least 5 has a $(\Delta + 1)$-certificate [8, Theorem 8].

So before Theorem 2.6, it was a genuine possibility that some non-bipartite regular graph, such as K_4 or one of Schauz's open graphs, has a clever Δ-certificate that is not of Galvin's form. Previous negative results were partial. McDonald [8, Theorem 5] ruled out certificates *of Galvin's form* for complete graphs, and our earlier notes ruled out all certificates for even complete graphs and for cubic graphs with a triangle. Theorem 2.6 rules out *every* orientation, for *every* non-bipartite regular graph. Theorem 16.3 then says that even on bipartite regular graphs there is nothing to find beyond Galvin's construction. Regularity is exactly the hypothesis that separates (iii) from the theorem, and Remark 6.3 shows the single place where it enters.

2.6 The proof in one paragraph, and the roadmap

Suppose D is a k-certificate for a simple k-regular graph G. Kernel-perfectness forces the k edges at each vertex u to be linearly ordered by D, so u ranks its neighbours $0, 1, \ldots, k-1$. The out-degree of an edge uv is then the rank of v at u plus the rank of u at v. Adding up over all edges shows that the budget $k-1$ is used exactly, so the two ranks of every edge add up to $k-1$. Hence if u puts v in its better half, v puts u in its worse half. Orienting each edge towards the better-half end gives an orientation of G (up to a perfect matching of "middle" edges when k is odd) in which every vertex has equal in-degree and out-degree. An odd cycle that follows this orientation would make D contain a directed odd cycle, which has no kernel. A parity argument shows that if G had any odd cycle at all, it would have one that follows the orientation. So G has no odd cycle, and is bipartite.

3 Preliminaries, all proved

This section collects every general fact the proof uses, each with a proof.

Step	Conclusion	Section
1	At each vertex the edges are linearly ordered; ranks $r_u(v) \in \{0, \dots, k-1\}$; $d^+(uv) = r_u(v) + r_v(u)$.	5
2	Tightness: $r_u(v) + r_v(u) = k - 1$ for every edge uv.	6
3	No odd ring: no odd cycle on which every vertex prefers its successor to its predecessor.	7
4	The digraph O and the set M_0: M_0 is empty (k even) or a perfect matching (k odd), and O is a balanced orientation of $G - M_0$.	8
5	Every admissible cycle is a ring; so there is no admissible odd cycle, and O has no directed closed walk of odd length.	9
6	Inside each component of $G - M_0$, any vertex reaches any other by a directed path of O.	10
7	Every component of $G - M_0$ is bipartite.	11
8	If k is even, G is bipartite.	12
9	If k is odd, G is bipartite.	13

Table 1: The nine steps of the proof of Theorem 2.6.

3.1 Two facts about numbers

Lemma 3.1. *For every integer $k \geq 1$, $0 + 1 + 2 + \cdots + (k-1) = \dfrac{k(k-1)}{2}$.*

Proof. Let $S = \sum_{i=0}^{k-1} i$. Writing the same sum backwards, $S = \sum_{i=0}^{k-1} (k-1-i)$. Adding the two expressions term by term, $2S = \sum_{i=0}^{k-1} \big(i + (k-1-i)\big) = \sum_{i=0}^{k-1} (k-1) = k(k-1)$. Divide by 2. □

Lemma 3.2 (a sum at its maximum)**.** *Let $a_1, \dots, a_N$ and c be real numbers with $a_j \leq c$ for every j. If $a_1 + \cdots + a_N = Nc$, then $a_j = c$ for every j.*

Proof. Each number $c - a_j$ is at least 0, and their sum is $Nc - (a_1 + \cdots + a_N) = 0$. A sum of non-negative numbers equals 0 only if every one of them equals 0. So $a_j = c$ for every j. □

3.2 Graphs

Definition 3.3 (simple graph)**.** A *simple graph* G consists of a finite set $V(G)$ of *vertices* and a set $E(G)$ of *edges*, each edge being a set $\{u, v\}$ of exactly two distinct vertices, its *ends*. We write uv for $\{u, v\}$, so $uv = vu$.

Because an edge has two *distinct* ends, there are no loops. Because $E(G)$ is a *set* of two-element sets, the edge uv occurs at most once, so there are no parallel edges. A *multigraph* is the more general object in which several edges may join the same two vertices; it is not covered by Theorem 2.6 (see Remark 5.4).

Lemma 3.4. *In a simple graph, two distinct edges have at most one common end.*

Proof. Suppose distinct edges e and f both have the ends u and v with $u \neq v$. An edge has exactly two ends, so $e = \{u, v\} = f$, contradicting $e \neq f$. □

For a vertex v we write $E(v)$ for the set of edges with v as an end, and $d(v) = |E(v)|$ for the *degree* of v. The vertices u with $uv \in E(G)$ are the *neighbours* of v; since G is simple, v has exactly $d(v)$ neighbours. G is *k-regular* if $d(v) = k$ for every vertex v. We write $n = |V(G)|$.

Lemma 3.5 (handshake)**.** $\sum_{v \in V(G)} d(v) = 2|E(G)|$. *In particular, if G is k-regular then $|E(G)| = nk/2$.*

Proof. Count the pairs (v, e) with v an end of e. Grouping by v gives $\sum_v d(v)$. Grouping by e gives 2 for every edge, since every edge has exactly two ends. So the two counts agree. □

Definition 3.6 (walks, paths, cycles)**.** A *walk* of *length* $L \geq 0$ from x_0 to x_L is a sequence of vertices $x_0, x_1, \ldots, x_L$ with $x_{i-1}x_i \in E(G)$ for $1 \leq i \leq L$. It is *closed* if $x_L = x_0$. A *path* is a walk whose vertices $x_0, \ldots, x_L$ are pairwise distinct. A *cycle* of length m is a closed walk $x_0, x_1, \ldots, x_m = x_0$ with $m \geq 3$ and $x_0, \ldots, x_{m-1}$ pairwise distinct. It is *odd* or *even* according to m.

We usually write a cycle as $a_1 a_2 \ldots a_m$, meaning the closed walk $a_1, \ldots, a_m, a_1$, with indices modulo m. Its edges are $a_i a_{i+1}$ for $1 \leq i \leq m$. These m edges are distinct: if $a_i a_{i+1} = a_j a_{j+1}$ with $i \neq j$, then, the vertices being distinct, $a_i = a_{j+1}$ and $a_{i+1} = a_j$, so $i \equiv j + 1$ and $i + 1 \equiv j \pmod{m}$, giving $2 \equiv 0 \pmod{m}$, impossible for $m \geq 3$.

Lemma 3.7 (odd closed walks contain odd cycles)**.** *Every closed walk of odd length in a simple graph contains an odd cycle. More precisely, the cycle uses only vertices and edges of the walk.*

Proof. By strong induction on the length L. Let $W = x_0, x_1, \ldots, x_L = x_0$ have odd length L.

(1) Suppose first that $x_0, \ldots, x_{L-1}$ are pairwise distinct. $L \neq 1$: a closed walk of length 1 would need the edge $x_0x_1 = x_0x_0$, which is not a two-element set. So $L \geq 3$, and W is a cycle by Definition 3.6. It is odd.

(2) Otherwise $x_i = x_j$ for some $0 \leq i < j \leq L - 1$. Split W into $W_1 = x_i, x_{i+1}, \ldots, x_j$ and $W_2 = x_j, x_{j+1}, \ldots, x_L, x_1, \ldots, x_i$. Both are closed walks, because $x_i = x_j$ and $x_L = x_0$. Their lengths are $j - i$ and $L - (j - i)$. Both lengths are at least 1 and at most $L - 1$.

(3) The two lengths add up to L, which is odd. So one of them is odd. By the induction hypothesis that walk, which is shorter than W and uses only vertices and edges of W, contains an odd cycle. □

Definition 3.8 (bipartite)**.** A *proper 2-colouring* of G is a map $\mathrm{col} : V(G) \to \{0, 1\}$ with $\mathrm{col}(u) \neq \mathrm{col}(v)$ for every edge uv. G is *bipartite* if it has one. Equivalently, $V(G)$ splits into two sets $A = \mathrm{col}^{-1}(0)$ and $B = \mathrm{col}^{-1}(1)$ with every edge having one end in each.

Lemma 3.9 (parity along a walk)**.** *Let* col *be a proper 2-colouring of G, and let W be a walk of length L from x to y. Then $L \equiv \mathrm{col}(x) + \mathrm{col}(y)$.*

Proof. By induction on L. If $L = 0$ then $x = y$ and $\mathrm{col}(x) + \mathrm{col}(y) = 2\,\mathrm{col}(x) \equiv 0 = L$. If $L \geq 1$, let $W = x_0, \ldots, x_L$. By induction $L - 1 \equiv \mathrm{col}(x_0) + \mathrm{col}(x_{L-1})$. The edge $x_{L-1}x_L$ has ends of different colours, so $\mathrm{col}(x_L) = 1 - \mathrm{col}(x_{L-1})$, that is $\mathrm{col}(x_L) \equiv \mathrm{col}(x_{L-1}) + 1$. Adding 1 to both sides of the induction hypothesis gives $L \equiv \mathrm{col}(x_0) + \mathrm{col}(x_{L-1}) + 1 \equiv \mathrm{col}(x_0) + \mathrm{col}(x_L)$. □

Definition 3.10 (components)**.** Two vertices x, y are *connected* if there is a walk from x to y. This is an equivalence relation: a walk of length 0 connects x to itself, walks can be reversed, and walks can be concatenated. Its classes are the *components*. We also write "component" for the subgraph formed by a class together with all edges having both ends in it. Every edge has both ends in the same component, since the edge itself is a walk.

Lemma 3.11 (bipartite means no odd cycle)**.** *A simple graph is bipartite if and only if it has no odd cycle.*

Proof. ($\Rightarrow$) Let col be a proper 2-colouring and let $a_1 \dots a_m$ be a cycle. It is a walk of length m from a_1 to a_1, so by Lemma 3.9 $m \equiv 2\,\mathrm{col}(a_1) \equiv 0$. Every cycle is even.

($\Leftarrow$) Suppose G has no odd cycle. By Lemma 3.7 it has no closed walk of odd length either. In each component choose one vertex, its *root*. For a vertex x with root ρ, let $\mathrm{col}(x)$ be the parity of the length of some walk from ρ to x; such a walk exists because x and ρ are in the same component.

(1) col *is well defined.* If W and W' are two walks from ρ to x, then W followed by W' reversed is a closed walk of length $|W| + |W'|$. It is even, so $|W| \equiv |W'|$.

(2) col *is proper.* Let xy be an edge. Both ends lie in the same component, with the same root ρ. If W is a walk from ρ to x, then W followed by the step x, y is a walk from ρ to y of length $|W| + 1$. So $\mathrm{col}(y) \equiv \mathrm{col}(x) + 1$, and $\mathrm{col}(x) \neq \mathrm{col}(y)$. □

Lemma 3.12. *A graph is bipartite if and only if each of its components is bipartite.*

Proof. ($\Rightarrow$) Restrict a proper 2-colouring to a component. ($\Leftarrow$) Take a proper 2-colouring of each component and combine them into one map on $V(G)$. Every edge has both ends in one component (Definition 3.10), where the colouring is proper. □

3.3 Digraphs

Definition 3.13 (digraph)**.** A *digraph* D consists of a finite set $V(D)$ and a set $A(D)$ of *arcs*, each arc an ordered pair (x, y) of *distinct* vertices, written $x \to y$. Its *tail* is x and its *head* is y. The *out-neighbours* of x are the y with $x \to y$; $N_D^+(x)$ is the set of them and $d_D^+(x) = |N_D^+(x)|$ is the *out-degree*. The *in-degree* $d_D^-(x)$ is the number of arcs with head x. For $S \subseteq V(D)$, the *induced subdigraph* $D[S]$ has vertex set S and the arcs of D with both ends in S.

Lemma 3.14. $\sum_{x \in V(D)} d_D^+(x) = |A(D)| = \sum_{x \in V(D)} d_D^-(x)$.

Proof. Every arc has exactly one tail and exactly one head. Counting arcs by their tails gives the first sum, and by their heads the second. □

Definition 3.15 (orientation)**.** Let H be a simple graph. An *orientation* of H is a digraph D with $V(D) = V(H)$ such that for every edge xy of H exactly one of $x \to y$ and $y \to x$ is an arc, and there are no other arcs. In an orientation, two vertices are joined by an arc iff they are adjacent in H, and then by exactly one arc. By Lemma 3.14, $\sum_x d_D^+(x) = |E(H)|$.

Definition 3.16 (directed walks and cycles)**.** A *directed walk* of length L from x_0 to x_L is a sequence $x_0, \dots, x_L$ with $x_{i-1} \to x_i$ an arc for $1 \leq i \leq L$. It is *closed* if $x_L = x_0$. A *directed path* is a directed walk with pairwise distinct vertices. A *directed cycle* of length m is a closed directed walk $x_0, \dots, x_m = x_0$ with $m \geq 2$ and $x_0, \dots, x_{m-1}$ pairwise distinct.

In an orientation of a simple graph there is no directed cycle of length 2, because that would need both $x \to y$ and $y \to x$. So in an orientation every directed cycle has length at least 3, its vertices are distinct, and its underlying vertices and edges form a cycle of the graph in the sense of Definition 3.6.

Lemma 3.17 (directed odd closed walks contain directed odd cycles)**.** *Every directed closed walk of odd length contains a directed cycle of odd length, using only vertices and arcs of the walk.*

Proof. Word for word the proof of Lemma 3.7, with "walk" replaced by "directed walk". Splitting a directed closed walk at a repeated vertex gives two directed closed walks, because each piece follows arcs forwards. The base case: a directed closed walk of length 1 would need an arc $x_0 \to x_0$, which is excluded by Definition 3.13; so an odd one with distinct vertices has length $L \geq 3$ and is a directed cycle. □

Lemma 3.18 (shortest walks are paths)**.** *If there is a directed walk from x to y, then there is a directed path from x to y, and it uses only vertices and arcs of the walk.*

Proof. Take a shortest directed walk $x = x_0, \dots, x_L = y$ among those using only vertices and arcs of the given walk. If $x_i = x_j$ with $i < j$, then deleting $x_{i+1}, \dots, x_j$ gives a shorter directed walk $x_0, \dots, x_i, x_{j+1}, \dots, x_L$ from x to y, still using only vertices and arcs of the given walk. That contradicts minimality, so the vertices are distinct. □

Definition 3.19 (balanced)**.** A digraph is *balanced* if $d^+(x) = d^-(x)$ for every vertex x.

Lemma 3.20 (arcs of balanced digraphs lie on cycles)**.** *Let D be a balanced digraph and let $x \to y$ be an arc. Then D has a directed path from y to x.*

Proof. Let R be the set of vertices z for which D has a directed walk from y to z. Then $y \in R$ (walk of length 0). Suppose, for a contradiction, that $x \notin R$.

(1) *No arc leaves R.* If $z \in R$ and $z \to w$, then a directed walk from y to z followed by the arc $z \to w$ is a directed walk from y to w, so $w \in R$.

(2) *At least one arc enters R,* namely $x \to y$, since $x \notin R$ and $y \in R$.

(3) *A count.* Consider $\Sigma = \sum_{z \in R} \big(d^+(z) - d^-(z)\big)$. Look at how a single arc $p \to q$ contributes to Σ. If $p, q \in R$, it adds 1 to $d^+(p)$ and 1 to $d^-(q)$, so it contributes $1 - 1 = 0$. If $p \in R$ and $q \notin R$ it contributes $+1$; there are no such arcs by (1). If $p \notin R$ and $q \in R$ it contributes -1. If $p, q \notin R$ it contributes 0. So Σ equals minus the number of arcs entering R, and by (2) $\Sigma \le -1$.

(4) But D is balanced, so every term $d^+(z) - d^-(z)$ is 0 and $\Sigma = 0$. This contradicts (3).

So $x \in R$: there is a directed walk from y to x, and by Lemma 3.18 a directed path. □

Lemma 3.21 (balanced digraphs are strongly connected inside components)**.** *Let D be a balanced digraph and let U be its underlying graph, which has an edge xy iff $x \to y$ or $y \to x$ is an arc. If x and y are in the same component H of U, then D has a directed path from x to y, and all its vertices lie in H.*

Proof. Take a walk $x = z_0, z_1, \dots, z_L = y$ in U. For each i, the edge $z_{i-1}z_i$ of U comes from an arc. If it is $z_{i-1} \to z_i$, use that arc. If it is $z_i \to z_{i-1}$, use the directed path from z_{i-1} to z_i given by Lemma 3.20. Concatenating these pieces gives a directed walk from x to y, and Lemma 3.18 turns it into a directed path. Every directed walk is in particular a walk of U, so it never leaves the component H of its starting vertex. □

3.4 Kernels, odd cycles and tournaments

We use Definition 2.2 for kernels and kernel-perfectness.

Lemma 3.22. *A kernel of a digraph with at least one vertex is non-empty.*

Proof. If $K = \emptyset$, then any vertex x is outside K, and (K2) asks for an arc from x into the empty set, which does not exist. □

Lemma 3.23 (a directed odd cycle has no kernel)**.** *Let $m \ge 3$ be odd, and let C be the digraph with vertices $x_1, \dots, x_m$ whose arcs are exactly $x_i \to x_{i+1}$ for $1 \le i \le m$ (indices modulo m). Then C has no kernel.*

Proof. Suppose K is a kernel. Note that x_i has exactly one out-neighbour, x_{i+1}.

(1) *If $x_i \in K$ then $x_{i+1} \notin K$.* The arc $x_i \to x_{i+1}$ joins them, and by (K1) no arc joins two vertices of K.

(2) *If $x_i \notin K$ then $x_{i+1} \in K$.* By (K2), x_i has an out-neighbour in K, and its only out-neighbour is x_{i+1}.

(3) Let $b_i = 1$ if $x_i \in K$ and $b_i = 0$ otherwise. By (1) and (2), $b_{i+1} = 1 - b_i$, so $b_{i+1} \equiv b_i + 1$, for every i.

(4) Applying (3) m times starting from x_1, $b_{1+m} \equiv b_1 + m$. But $x_{1+m} = x_1$, so $b_1 \equiv b_1 + m$, that is $m \equiv 0$. This contradicts m odd. □

A *tournament* is an orientation of a complete graph: between any two distinct vertices there is exactly one arc.

Lemma 3.24 (kernels of tournaments are sinks)**.** *If T is a tournament with at least one vertex and K is a kernel of T, then $K = \{s\}$ for a single vertex s, and every other vertex x has the arc $x \to s$.*

Proof. Any two distinct vertices of T are joined by an arc, so by (K1) K has at most one vertex. By Lemma 3.22 it has at least one. So $K = \{s\}$, and (K2) says that every $x \neq s$ has an arc into K, that is, $x \to s$. □

Lemma 3.25 (kernel-perfect tournaments are transitive)**.** *Let T be a tournament such that $T[S]$ has a kernel for every set S of vertices. If $x \to y$ and $y \to z$, then $x \neq z$ and $x \to z$.*

Proof. If $x = z$ then $x \to y$ and $y \to x$ are both arcs, which a tournament does not allow. So x, y, z are three distinct vertices, and there is exactly one arc between x and z. If it were $z \to x$, then $T[\{x, y, z\}]$ would have exactly the arcs $x \to y$, $y \to z$, $z \to x$. That is the digraph of Lemma 3.23 with $m = 3$, which has no kernel, contrary to the hypothesis. Hence the arc is $x \to z$. □

Lemma 3.26 (ranks in a transitive tournament)**.** *Let T be a tournament on $d \geq 1$ vertices with the property of Lemma 3.25. For each vertex x put $\rho(x) = d_T^+(x)$. Then*

(a) *for distinct x, y: $x \to y$ iff $\rho(x) > \rho(y)$;*

(b) *ρ is a bijection from $V(T)$ onto $\{0, 1, \ldots, d-1\}$.*

Proof. **(1)** *If $x \to y$ then $N^+(y) \subseteq N^+(x)$.* Let $z \in N^+(y)$, so $y \to z$. By Lemma 3.25, $x \neq z$ and $x \to z$, so $z \in N^+(x)$.

(2) *If $x \to y$ then $\rho(x) > \rho(y)$.* By (1), $N^+(y) \subseteq N^+(x)$. Also $y \in N^+(x)$ but $y \notin N^+(y)$, since there are no arcs from a vertex to itself. So $N^+(x)$ contains $N^+(y)$ and at least one more element, and $\rho(x) \geq \rho(y) + 1$.

(3) *Converse.* Let $x \neq y$ with $\rho(x) > \rho(y)$. There is an arc between them. If it were $y \to x$, then (2) would give $\rho(y) > \rho(x)$. So it is $x \to y$. This proves (a).

(4) *ρ is injective.* For distinct x, y one of the arcs $x \to y$, $y \to x$ exists, and by (2) $\rho(x) \neq \rho(y)$.

(5) *ρ is onto $\{0, \ldots, d-1\}$.* A vertex has at most $d-1$ out-neighbours, so ρ takes values in this set of d numbers. An injective map from a set of d elements into a set of d elements is a bijection. This proves (b). □

4 The setting

Definition 4.1 (line graph)**.** The *line graph* $L(G)$ of a simple graph G has vertex set $E(G)$, and two distinct edges e, f of G are adjacent in $L(G)$ iff they have a common end.

Lemma 4.2 (stars are cliques)**.** *Let G be simple.*

(a) *For each vertex w, any two distinct edges in $E(w)$ are adjacent in $L(G)$.*

(b) *If e, f are adjacent in $L(G)$, they have exactly one common end w, and then $e, f \in E(w)$ and e, f are in no other common $E(w')$.*

(c) $|E(L(G))| = \sum_{w \in V(G)} \binom{d(w)}{2}$.

Proof. (a) Both contain w. (b) They have at least one common end by definition, and at most one by Lemma 3.4. Being in $E(w')$ means having w' as an end, so w' is a common end, and $w' = w$. (c) By (a) and (b), the map sending an edge $\{e, f\}$ of $L(G)$ to the pair $\big(w, \{e, f\}\big)$, where w is the common end, is a bijection onto the set of pairs $\big(w, \{e, f\}\big)$ with $\{e, f\}$ a two-element subset of $E(w)$. For a fixed w there are $\binom{d(w)}{2}$ such subsets. □

From now on, until the end of Section 13, we work under the following hypothesis.

Standing Hypothesis 4.3. $k \geq 1$ is an integer; G is a simple k-regular graph with n vertices; D is a k-certificate for G (Definition 2.4). Explicitly: D is an orientation of $L(G)$, D is kernel-perfect, and $d^+_D(e) \leq k - 1$ for every edge e of G.

We write $e \to f$ for an arc of D between two edges e, f of G, and $d^+(e)$ for $d^+_D(e)$.

Remark 4.4 (which hypotheses each step uses). Step 1 uses only that G is simple and that D is a kernel-perfect orientation of $L(G)$. Step 2 is the only step that uses regularity and the budget $k - 1$. Step 3 uses kernel-perfectness again. Steps 4 to 9 use only the conclusions of Steps 1 to 3 and the simplicity of G.

5 Step 1: ranks

Lemma 5.1 (star lemma)**.** *Let G be simple and D a kernel-perfect orientation of $L(G)$. For every vertex w of G with $d(w) \geq 1$, the subdigraph $D[E(w)]$ is a tournament with the property of Lemma 3.25.*

Proof. By Lemma 4.2(a), any two distinct edges in $E(w)$ are adjacent in $L(G)$, and D is an orientation, so exactly one arc joins them. Hence $D[E(w)]$ is a tournament. For any $S \subseteq E(w)$, the digraph $D[E(w)][S]$ equals $D[S]$, which has a kernel because D is kernel-perfect. □

Definition 5.2 (ranks)**.** For a vertex w and an edge $e \in E(w)$, the *rank of e at w* is

$$r_w(e) = \big|\{\, f \in E(w) : e \to f \,\}\big|,$$

the out-degree of e inside the tournament $D[E(w)]$. For an edge wv we also write $r_w(v) = r_w(wv)$, the *rank of v at w*. Rank 0 is the best rank. We say *w prefers v to x* if $r_w(v) < r_w(x)$.

So every edge uv carries two ranks: $r_u(v)$, the rank u gives it, and $r_v(u)$, the rank v gives it.

Lemma 5.3 (properties of ranks)**.** *Assume G simple and D a kernel-perfect orientation of $L(G)$. Let w be a vertex with $d(w) \geq 1$.*

(a) *r_w is a bijection from $E(w)$ onto $\{0, 1, \ldots, d(w)-1\}$. Equivalently, $v \mapsto r_w(v)$ is a bijection from the neighbours of w onto $\{0, \ldots, d(w) - 1\}$.*

(b) *For distinct $e, f \in E(w)$: $e \to f$ iff $r_w(e) > r_w(f)$. In words: an edge points to the edges that are ranked better than it at a common end.*

(c) *For every edge $e = uv$: $d^+(e) = r_u(e) + r_v(e) = r_u(v) + r_v(u)$.*

Proof. (a) and (b) are Lemma 3.26, applied to the tournament $D[E(w)]$ with $d = d(w)$, which is allowed by Lemma 5.1. The rank $r_w(e)$ is exactly $\rho(e)$ there. The second sentence of (a) holds because $v \mapsto wv$ is a bijection from the neighbours of w onto $E(w)$.

(c) The out-neighbours of e in D are among its neighbours in $L(G)$, which are the edges $f \neq e$ with a common end with e. The ends of e are u and v, so these are the edges of $\big(E(u) \cup E(v)\big) \setminus \{e\}$.

(1) $E(u) \cap E(v) = \{e\}$: an edge in both has ends u and v, so it is e by Lemma 3.4 (or simply because $\{u, v\}$ is one set).

(2) Hence the out-neighbours of e split into two disjoint sets: those in $E(u)$ and those in $E(v)$.

(3) The first set is $\{f \in E(u) : e \to f\}$, of size $r_u(e)$. The second is $\{f \in E(v) : e \to f\}$, of size $r_v(e)$.

(4) So $d^+(e) = r_u(e) + r_v(e)$. By the notation of Definition 5.2, $r_u(e) = r_u(v)$ and $r_v(e) = r_v(u)$. □

Remark 5.4 (where simplicity is used). Step (1) in the proof of (c) is where simplicity matters. If G had two parallel edges e, e' joining u and v, then e' would lie in both $E(u)$ and $E(v)$, and the single arc between e and e' would have to agree with the order at u and with the order at v at the same time. The out-degree would no longer split as a sum of two independent ranks, and Step 2 below would not follow as written. This is why multigraphs are excluded.

Remark 5.5. Lemma 5.1 is not new: it is the necessity of the "every clique has a kernel" condition in the characterisation of kernel-perfect orientations of line graphs by Borodin, Kostochka and Woodall [3]. The proof above is self-contained.

Under Standing Hypothesis 4.3, $d(w) = k$ for every w, so:

> **Conclusion of Step 1.** Each vertex w ranks its k neighbours with the numbers $0, 1, \dots, k-1$, each used once. For distinct $e, f \in E(w)$, $e \to f$ iff $r_w(e) > r_w(f)$. For every edge uv, $d^+(uv) = r_u(v) + r_v(u)$.

6 Step 2: the budget is used exactly

Lemma 6.1 (tightness)**.** *Under Standing Hypothesis 4.3, $d^+(e) = k - 1$ for every edge e of G, and*

$$r_u(v) + r_v(u) = k - 1 \qquad \text{for every edge } uv \text{ of } G.$$

Proof. **(1)** D is an orientation of $L(G)$, so by Definition 3.15, $\sum_{e \in E(G)} d^+(e) = |E(L(G))|$.

(2) By Lemma 4.2(c) and regularity, $|E(L(G))| = \sum_w \binom{k}{2} = n \cdot \dfrac{k(k-1)}{2}$.

(3) By Lemma 3.5, $|E(G)| = nk/2$, so $n \cdot \dfrac{k(k-1)}{2} = \dfrac{nk}{2} \cdot (k-1) = |E(G)|\,(k-1)$.

(4) By (1), (2) and (3), $\sum_{e \in E(G)} d^+(e) = |E(G)|\,(k-1)$.

(5) By Standing Hypothesis 4.3, every one of the $|E(G)|$ terms satisfies $d^+(e) \leq k - 1$.

(6) By Lemma 3.2, with $N = |E(G)|$, $a_e = d^+(e)$ and $c = k - 1$, every $d^+(e)$ equals $k - 1$.

(7) By Lemma 5.3(c), $r_u(v) + r_v(u) = d^+(uv) = k - 1$. □

Remark 6.2 (a second count, as a cross-check). The same total can be computed from the ranks. By Lemma 5.3(c), $\sum_e d^+(e) = \sum_{uv \in E} \big(r_u(v) + r_v(u)\big)$. Every term $r_w(x)$ with $wx \in E$ appears exactly once on the right, so this equals $\sum_w \sum_x r_w(x)$, and by Lemma 5.3(a) and Lemma 3.1 the inner sum is $0 + 1 + \cdots + (k-1) = k(k-1)/2$ for every w. The total is $nk(k-1)/2$, as in step (2).

Remark 6.3 (this is the only use of regularity). For an arbitrary simple graph G of maximum degree Δ and a Δ-certificate, the same computation gives

$$|E(G)|\,(\Delta-1)-\sum_{e} d^{+}(e) \;=\; \sum_{uv\in E(G)} \Big(\Delta-\frac{d(u)+d(v)}{2}\Big),$$

because $\sum_w \binom{d(w)}{2} = \sum_w \sum_{e\in E(w)} \frac{d(w)-1}{2} = \sum_{uv\in E} \big(\frac{d(u)-1}{2}+\frac{d(v)-1}{2}\big)$. The right side is the total *slack* available to the orientation. It is 0 exactly when every edge has both ends of degree Δ. For a regular graph there is no slack, which forces tightness. For a non-regular graph the slack can be spent, and Theorem 2.6 genuinely fails: the diamond has slack $\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+0=2$ and does have 3-certificates (Table 2).

Conclusion of Step 2. $r_u(v)+r_v(u)=k-1$ for every edge uv.

7 Step 3: there is no odd ring

Definition 7.1 (ring)**.** A cycle $a_1a_2\dots a_m$ of G, read in this direction, is a *ring* if every vertex on it prefers the next vertex to the previous one:

$$r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1}) \qquad \text{for every } i, \text{ indices modulo } m.$$

It is an *odd ring* if m is odd.

Lemma 7.2 (no odd ring)**.** *Let G be simple and D a kernel-perfect orientation of $L(G)$. Then G has no odd ring.*

Proof. Suppose $a_1\dots a_m$ is an odd ring, $m\ge 3$ odd. Put $e_i=a_ia_{i+1}$ for $1\le i\le m$, so that $e_{i-1}=a_{i-1}a_i$, and let $S=\{e_1,\dots,e_m\}$.

(1) *The e_i are m distinct edges.* They are the edges of a cycle (see the remark after Definition 3.6).

(2) *For each i, $e_{i-1}\to e_i$.* Both $e_{i-1}=a_{i-1}a_i$ and $e_i=a_ia_{i+1}$ lie in $E(a_i)$. Their ranks at a_i are $r_{a_i}(a_{i-1})$ and $r_{a_i}(a_{i+1})$, and the ring condition says $r_{a_i}(a_{i-1})>r_{a_i}(a_{i+1})$. By Lemma 5.3(b), $e_{i-1}\to e_i$.

(3) *Non-consecutive edges are not adjacent.* Let $j\notin\{i-1,i,i+1\}$ modulo m. The ends of e_i are a_i,a_{i+1} and those of e_j are a_j,a_{j+1}. The vertices $a_1,\dots,a_m$ are distinct, so a common end would need $i\equiv j$, $i\equiv j+1$, $i+1\equiv j$ or $i+1\equiv j+1$, that is $j\in\{i-1,i,i+1\}$. So e_i,e_j have no common end and are not adjacent in $L(G)$, and D has no arc between them.

(4) *Each consecutive pair is counted once.* If $\{e_{i-1},e_i\}=\{e_{j-1},e_j\}$ with $i\not\equiv j$, then $e_{i-1}=e_j$ and $e_i=e_{j-1}$, so by (1) $i-1\equiv j$ and $i\equiv j-1$, giving $2\equiv 0 \pmod m$, impossible for $m\ge 3$.

(5) *So $D[S]$ is a directed odd cycle.* By (2), (3) and (4), the arcs of $D[S]$ are exactly $e_{i-1}\to e_i$ for $1\le i\le m$; between consecutive edges D has exactly one arc because it is an orientation, and (2) says which. Writing $x_i=e_i$, this is the digraph of Lemma 3.23.

(6) By Lemma 3.23, $D[S]$ has no kernel. But D is kernel-perfect. Contradiction. □

Remark 7.3. For $m=3$ every two of e_1,e_2,e_3 are consecutive, so step (3) is vacuous and $D[S]$ is a directed triangle. This recovers the triangle obstruction of our earlier note on cubic graphs.

Conclusion of Step 3. G has no odd ring.

8 Step 4: an orientation of G itself

From here on the certificate D is no longer used directly. Everything follows from the conclusions of Steps 1, 2 and 3.

Definition 8.1 (the shifted rank y)**.** For every edge uv of G and each of its two ends, put

$$y(u,v) = r_u(v) - \tfrac{k-1}{2}, \qquad y(v,u) = r_v(u) - \tfrac{k-1}{2}.$$

So $y(u,v) < 0$ means that u puts v in the better half of its ranking, $y(u,v) > 0$ means the worse half, and $y(u,v) = 0$ means exactly the middle rank $\frac{k-1}{2}$, which exists only when k is odd.

Lemma 8.2 (antisymmetry)**.** *$y(v,u) = -y(u,v)$ for every edge uv.*

Proof. $y(u,v) + y(v,u) = r_u(v) + r_v(u) - (k-1) = 0$ by Lemma 6.1. □

Definition 8.3 (M_0 and O)**.** Let $M_0 = \{uv \in E(G) : y(u,v) = 0\}$. By Lemma 8.2 this does not depend on which end is written first. Let O be the digraph with vertex set $V(G)$ and an arc $u \to v$ for every edge $uv \notin M_0$ with $y(u,v) < 0$. We use $\to$ for arcs of O from now on; arcs of D no longer appear.

Lemma 8.4. *O is an orientation of the graph $G - M_0$ (vertex set $V(G)$, edge set $E(G) \setminus M_0$).*

Proof. Let $uv \in E(G) \setminus M_0$. Then $y(u,v) \neq 0$, and by Lemma 8.2 $y(v,u) = -y(u,v)$. So exactly one of $y(u,v)$ and $y(v,u)$ is negative, and exactly one of $u \to v$, $v \to u$ is an arc. O has no other arcs by definition. □

Lemma 8.5 (the middle edges)**.** (a) *If k is even, $M_0 = \emptyset$.*

(b) *If k is odd, M_0 is a perfect matching: every vertex is an end of exactly one edge of M_0. In particular two distinct edges of M_0 have no common end.*

Proof. (a) $y(u,v) = 0$ means $r_u(v) = \frac{k-1}{2}$. If k is even, $\frac{k-1}{2}$ is not an integer and ranks are integers.

(b) Let k be odd and fix a vertex u. An edge uv is in M_0 iff $r_u(v) = \frac{k-1}{2}$. Since $\frac{k-1}{2}$ is one of $0, \ldots, k-1$ and r_u is a bijection onto these numbers (Lemma 5.3(a)), there is exactly one such v. So u is an end of exactly one edge of M_0. If two distinct edges of M_0 had a common end u, then u would be an end of two edges of M_0. □

Lemma 8.6 (O is balanced)**.** *For every vertex u, $d^+_O(u) = d^-_O(u) = \lfloor k/2 \rfloor$.*

Proof. **(1)** The out-arcs of u in O go to the neighbours v with $y(u,v) < 0$, that is $r_u(v) < \frac{k-1}{2}$.

(2) An arc $v \to u$ means $y(v,u) < 0$, that is $y(u,v) > 0$ by Lemma 8.2, that is $r_u(v) > \frac{k-1}{2}$. So the in-arcs of u come from the neighbours with $r_u(v) > \frac{k-1}{2}$.

(3) By Lemma 5.3(a) the ranks $r_u(v)$ are $0, 1, \ldots, k-1$, each once. If k is even, $k = 2t$, the ranks below $\frac{k-1}{2} = t - \frac{1}{2}$ are $0, \ldots, t-1$ and those above are $t, \ldots, 2t-1$: t each. If k is odd, $k = 2t+1$, the ranks below $\frac{k-1}{2} = t$ are $0, \ldots, t-1$ and those above are $t+1, \ldots, 2t$: t each. In both cases $t = \lfloor k/2 \rfloor$. □

Conclusion of Step 4. M_0 is empty (k even) or a perfect matching (k odd). O is a balanced orientation of $G - M_0$, with $u \to v$ iff $y(u,v) < 0$.

9 Step 5: admissible cycles are rings

Definition 9.1 (admissible)**.** A cycle $a_1 \dots a_m$ of G, read in this direction, is *admissible* if for every i either $a_i a_{i+1} \in M_0$ or $a_i \to a_{i+1}$ is an arc of O.

Lemma 9.2. *A cycle $a_1 \dots a_m$ is admissible iff $y(a_i, a_{i+1}) \leq 0$ for every i.*

Proof. For an edge in M_0, $y(a_i, a_{i+1}) = 0$. For an edge not in M_0, $a_i \to a_{i+1}$ iff $y(a_i, a_{i+1}) < 0$, by Definition 8.3. So "in M_0 or forwards in O" is the same as "$y(a_i, a_{i+1}) = 0$ or $y(a_i, a_{i+1}) < 0$". □

Lemma 9.3 (admissible cycles are rings)**.** *Every admissible cycle is a ring.*

Proof. Let $a_1 \dots a_m$ be admissible and fix i. We must show $r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1})$.

(1) By Definition 8.1, $r_{a_i}(a_{i+1}) - r_{a_i}(a_{i-1}) = y(a_i, a_{i+1}) - y(a_i, a_{i-1})$, since the $\frac{k-1}{2}$ cancels.

(2) By Lemma 8.2, $-y(a_i, a_{i-1}) = y(a_{i-1}, a_i)$. So the difference is $y(a_{i-1}, a_i) + y(a_i, a_{i+1})$.

(3) Both terms are ≤ 0, by Lemma 9.2 applied at $i-1$ and at i.

(4) They are not both 0. If they were, both edges $a_{i-1}a_i$ and $a_i a_{i+1}$ would be in M_0. These are two distinct edges (edges of a cycle, Definition 3.6) with the common end a_i. If k is even, M_0 is empty. If k is odd, two distinct edges of M_0 have no common end (Lemma 8.5(b)). Either way this is impossible.

(5) So the sum is strictly negative, and $r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1})$. □

Corollary 9.4. *G has no admissible odd cycle.*

Proof. It would be an odd ring (Lemma 9.3), and there is none (Lemma 7.2). □

Corollary 9.5. *O has no directed cycle of odd length, and no directed closed walk of odd length.*

Proof. A directed cycle of O of odd length m has $m \geq 3$ and its vertices $a_1, \dots, a_m$ form a cycle of G (remark after Definition 3.16), read with every edge forwards in O. That is an admissible odd cycle, which Corollary 9.4 excludes. A directed closed walk of odd length would contain a directed odd cycle by Lemma 3.17. □

Conclusion of Step 5. No admissible odd cycle exists. In particular O has no directed closed walk of odd length.

10 Step 6: moving around inside a component

Lemma 10.1. *Let H be a component of $G - M_0$, and let $x, y \in V(H)$. Then O has a directed path from x to y, all of whose vertices lie in H. Moreover, for every arc $x \to y$ of O, there is a directed path of O from y to x.*

Proof. O is balanced (Lemma 8.6), and its underlying graph is $G - M_0$ (Lemma 8.4). Apply Lemma 3.21 for the first claim and Lemma 3.20 for the second. □

Conclusion of Step 6. Inside each component of $G - M_0$, O can get from any vertex to any other along a directed path. Every arc of O has a directed return path.

11 Step 7: every component of $G - M_0$ is bipartite

Lemma 11.1. *Every component H of $G - M_0$ is bipartite.*

Proof. Suppose H is not bipartite. By Lemma 3.11, H has an odd cycle $z_0 z_1 \dots z_{m-1}$, m odd, indices modulo m. Each edge $z_i z_{i+1}$ is an edge of $G - M_0$, so by Lemma 8.4 it carries exactly one arc of O: either $z_i \to z_{i+1}$ ("forwards") or $z_{i+1} \to z_i$ ("backwards"). We build a directed closed walk W of O by going once around the cycle and replacing each edge by a piece.

(1) *Forwards edge.* If $z_i \to z_{i+1}$, the piece Q_i is that single arc. It is a directed walk from z_i to z_{i+1} of length 1.

(2) *Backwards edge.* If $z_{i+1} \to z_i$, then by Lemma 10.1 there is a directed path Q_i of O from z_i to z_{i+1} (the return path of that arc).

(3) *Every piece has odd length.* In case (1) the length is 1. In case (2), the arc $z_{i+1} \to z_i$ followed by Q_i is a directed closed walk of length $1 + |Q_i|$. By Corollary 9.5 this length is even, so $|Q_i|$ is odd.

(4) *The pieces join up.* Q_i ends at z_{i+1}, where Q_{i+1} starts. So $W = Q_0 Q_1 \cdots Q_{m-1}$ is a directed walk from z_0 to $z_m = z_0$, that is, a directed closed walk.

(5) *W has odd length.* Its length is a sum of m odd numbers, and m is odd, so the length is odd.

(6) This contradicts Corollary 9.5. Hence H is bipartite. □

Conclusion of Step 7. Every component of $G - M_0$ is bipartite.

12 Step 8: the case k even

Lemma 12.1. *If k is even, G is bipartite.*

Proof. By Lemma 8.5(a), $M_0 = \emptyset$, so $G - M_0 = G$. By Lemma 11.1, every component of G is bipartite, and by Lemma 3.12 so is G. □

Conclusion of Step 8. If k is even, G is bipartite.

13 Step 9: the case k odd

Why this step is needed. When k is odd, $G - M_0$ can be bipartite while G is not. The smallest example is K_4 in Example 15.2: $G - M_0$ is a 4-cycle, and the two edges of M_0 are its diagonals, each joining two vertices of the same colour. So we must show that the matching edges cannot spoil bipartiteness without creating an admissible odd cycle.

If $k = 1$, then G is 1-regular, so G is a perfect matching, which is bipartite (colour the two ends of each edge 0 and 1). **From now on in this section $k \geq 3$ is odd.**

Set-up. By Lemma 11.1 each component of $G - M_0$ has a proper 2-colouring. Combine them into one map $\mathrm{col} : V(G) \to \{0, 1\}$, which is a proper 2-colouring of $G - M_0$ (Lemma 3.12). For a vertex x let $\mathrm{comp}(x)$ be the component of $G - M_0$ containing x. For an edge $f = uv \in M_0$ define its *label*

$$\lambda(f) = \begin{cases} 1 & \text{if } \mathrm{col}(u) = \mathrm{col}(v), \\ 0 & \text{if } \mathrm{col}(u) \neq \mathrm{col}(v). \end{cases}$$

Since $\mathrm{col}(u) + \mathrm{col}(v)$ is even exactly when $\mathrm{col}(u) = \mathrm{col}(v)$, we have, for every $f = uv \in M_0$,

$$\lambda(f) \equiv \mathrm{col}(u) + \mathrm{col}(v) + 1. \tag{13.1}$$

An edge of M_0 with $\lambda = 1$ is exactly a "bad" edge for col: its two ends have the same colour.

Definition 13.1 (jumps and jump sequences)**.** A *jump* is an ordered pair (v, u) of vertices with $vu \in M_0$; we say it *leaves* $\mathrm{comp}(v)$ from v and *enters* $\mathrm{comp}(u)$ at u. Its label is $\lambda(vu)$. A *jump sequence* is a finite list of jumps $(v_1, u_2), (v_2, u_3), \ldots, (v_s, u_{s+1})$, with $s \geq 0$, such that each jump leaves from the component the previous jump entered: $\mathrm{comp}(u_i) = \mathrm{comp}(v_i)$ for $2 \leq i \leq s$. It goes from $\mathrm{comp}(v_1)$ to $\mathrm{comp}(u_{s+1})$ (if $s = 0$ it goes from any component to itself). It is a *jump cycle* of length $t = s \geq 1$ if moreover $\mathrm{comp}(u_{t+1}) = \mathrm{comp}(v_1)$; then we write $u_1 = u_{t+1}$ and read indices modulo t. The *label sum* Λ of a jump sequence is the sum of the labels of its jumps.

The *reverse* of a jump (v, u) is (u, v); it has the same label. The reverse of a jump sequence (reverse each jump and their order) is a jump sequence going the other way, with the same label sum.

Lemma 13.2 (even jump cycles give a 2-colouring)**.** *If every jump cycle has even label sum, then G is bipartite.*

Proof. Call two components *linked* if there is a jump sequence from one to the other. This is an equivalence relation: the empty sequence links a component to itself, sequences can be reversed, and sequences can be concatenated (the last jump of the first enters some component, and the first jump of the second leaves from that same component). In each class choose a component, its *root*. For a component H with root R, let $\varphi(H) \in \{0, 1\}$ be the parity of the label sum of some jump sequence from R to H.

(1) *φ is well defined.* Let J and J' be jump sequences from R to H. If both are empty, both label sums are 0. Otherwise J followed by the reverse of J' is a jump cycle: it leaves R, travels to H, and returns to R. Its label sum $\Lambda(J) + \Lambda(J')$ is even by hypothesis, so $\Lambda(J) \equiv \Lambda(J')$.

(2) *The relation across a matching edge.* Let $xy \in M_0$, $H = \mathrm{comp}(x)$, $H' = \mathrm{comp}(y)$. Take a jump sequence J from the root R to H, and append the jump (x, y). This is allowed, since J enters $H = \mathrm{comp}(x)$ (or J is empty and starts in H). The result goes from R to H', so H' has the same root, and $\varphi(H') \equiv \varphi(H) + \lambda(xy)$.

(3) *The new colouring.* Define $\mathrm{col}'(x) \equiv \mathrm{col}(x) + \varphi(\mathrm{comp}(x))$, so col' is col with the colours swapped on every component H with $\varphi(H) = 1$.

(4) col' *is proper on edges not in M_0.* Such an edge xy lies in one component, so $\varphi(\mathrm{comp}(x)) = \varphi(\mathrm{comp}(y))$, and $\mathrm{col}(x) \neq \mathrm{col}(y)$. Adding the same bit to both keeps them different.

(5) col' *is proper on edges in M_0.* For $xy \in M_0$, using (2) and (13.1),

$$\begin{aligned} \mathrm{col}'(x) + \mathrm{col}'(y) &\equiv \mathrm{col}(x) + \mathrm{col}(y) + \varphi(H) + \varphi(H') \\ &\equiv \mathrm{col}(x) + \mathrm{col}(y) + \lambda(xy) \\ &\equiv 2\,\mathrm{col}(x) + 2\,\mathrm{col}(y) + 1 \equiv 1. \end{aligned}$$

So $\mathrm{col}'(x) \neq \mathrm{col}'(y)$.

(6) By (4) and (5), col' is a proper 2-colouring of G. □

Lemma 13.3 (a shortest odd jump cycle)**.** *Suppose G is not bipartite. Then there is a jump cycle with odd label sum. Let $(v_1, u_2), \ldots, (v_t, u_1)$ be one with the smallest possible length t, write $f_i = v_i u_{i+1}$ and $H_i = \mathrm{comp}(u_i) = \mathrm{comp}(v_i)$. Then:*

(a) *the components $H_1, \ldots, H_t$ are pairwise distinct;*

(b) *$u_i \neq v_i$ for every i.*

Proof. The first sentence is the contrapositive of Lemma 13.2.

(a) Suppose $H_i = H_j$ with $1 \le i < j \le t$. Split the jump cycle into

$$C_1 : (v_i, u_{i+1}), \dots, (v_{j-1}, u_j) \qquad \text{and} \qquad C_2 : (v_j, u_{j+1}), \dots, (v_t, u_1), (v_1, u_2), \dots, (v_{i-1}, u_i).$$

(1) C_1 is a jump cycle: consecutive jumps match as in the original, and it closes up because its last jump enters $\mathrm{comp}(u_j) = H_j = H_i = \mathrm{comp}(v_i)$, where its first jump leaves. Its length is $j - i$.

(2) C_2 is a jump cycle for the same reason, closing up because $\mathrm{comp}(u_i) = H_i = H_j = \mathrm{comp}(v_j)$. Its length is $t - (j - i)$.

(3) Both lengths are between 1 and $t - 1$.

(4) $\Lambda(C_1) + \Lambda(C_2)$ is the label sum of the original cycle, which is odd, so one of $\Lambda(C_1), \Lambda(C_2)$ is odd. That jump cycle is shorter than t, contradicting the choice of t.

(b) If $t = 1$, the only jump is (v_1, u_1) and $f_1 = v_1u_1$ is an edge, so $v_1 \neq u_1$ by Definition 3.3. Let $t \ge 2$ and fix i. The vertex u_i is an end of $f_{i-1} = v_{i-1}u_i$ and v_i is an end of $f_i = v_iu_{i+1}$ (indices modulo t).

(1) $f_{i-1} \neq f_i$ *when* $t \ge 3$. The ends of f_{i-1} lie in H_{i-1} and H_i, and the ends of f_i lie in H_i and H_{i+1}. If $f_{i-1} = f_i$, the two ends would lie in the same pair of components, so $\{H_{i-1}, H_i\} = \{H_i, H_{i+1}\}$. Since $H_{i-1} \neq H_i$ by (a), this forces $H_{i-1} = H_{i+1}$, which contradicts (a) because $i - 1 \not\equiv i + 1 \pmod t$ for $t \ge 3$.

(2) $f_{i-1} \neq f_i$ *when* $t = 2$. Then $\{f_{i-1}, f_i\} = \{f_1, f_2\}$, and if $f_1 = f_2$ the label sum would be $2\lambda(f_1)$, which is even. But it is odd.

(3) So f_{i-1} and f_i are two distinct edges of the matching M_0, and by Lemma 8.5(b) they have no common end. Hence $u_i \neq v_i$. □

Lemma 13.4. *If k is odd, G is bipartite.*

Proof. The case $k = 1$ was handled above, so let $k \ge 3$. Suppose G is not bipartite, and take the jump cycle $(v_1, u_2), \dots, (v_t, u_1)$ of Lemma 13.3, with $f_i = v_iu_{i+1}$ and components $H_1, \dots, H_t$.

(1) *Paths inside the components.* For each i, u_i and v_i lie in H_i, so by Lemma 10.1 there is a directed path P_i of O from u_i to v_i with all vertices in H_i. Since $u_i \neq v_i$ (Lemma 13.3(b)), $|P_i| \ge 1$.

(2) *The closed walk Z.* Let Z be: follow P_1 from u_1 to v_1, take the edge f_1 to u_2, follow P_2 to v_2, take f_2 to u_3, and so on, ending with f_t from v_t back to u_1. Consecutive pieces meet correctly, so Z is a closed walk of G of length $|Z| = \sum_{i=1}^{t} |P_i| + t$.

(3) *The vertices of Z are distinct.* Listing Z from u_1 and stopping before the return, its vertices are those of P_1, then of P_2, and so on. Within P_i they are distinct because P_i is a path. Between different i they are distinct because they lie in the components H_i, which are pairwise distinct (Lemma 13.3(a)) and therefore disjoint. The number of vertices listed is $\sum_i (|P_i| + 1) = |Z|$.

(4) $|Z|$ *is odd.* The path P_i is a walk in $G - M_0$, where col is proper, so Lemma 3.9 gives $|P_i| \equiv \mathrm{col}(u_i) + \mathrm{col}(v_i)$. Hence

$$|Z| \equiv \sum_{i=1}^{t} \big(\mathrm{col}(u_i) + \mathrm{col}(v_i)\big) + t.$$

As i runs over $1, \ldots, t$, so does $i+1$ modulo t; hence $\sum_i \mathrm{col}(u_i) = \sum_i \mathrm{col}(u_{i+1})$, and

$$|Z| \equiv \sum_{i=1}^{t} \big(\mathrm{col}(v_i) + \mathrm{col}(u_{i+1}) + 1\big) \equiv \sum_{i=1}^{t} \lambda(f_i) = \Lambda \equiv 1,$$

using (13.1) for $f_i = v_i u_{i+1}$.

(5) *Z is a cycle.* $|Z| \geq t + t \geq 2$ by (1), and $|Z|$ is odd by (4), so $|Z| \geq 3$. Together with (3), Z is a cycle in the sense of Definition 3.6, and an odd one.

(6) *Z is admissible*, read in the direction just described. Each edge of Z is either an arc of some P_i, traversed forwards in O, or one of the f_i, which lie in M_0. This is Definition 9.1.

(7) This contradicts Corollary 9.4. So G is bipartite. □

Conclusion of Step 9. If k is odd, G is bipartite.

14 Proof of the main theorem

Proof of Theorem 2.6. Let G be a simple k-regular graph with $k \geq 1$, and let D be a k-certificate for G. Then Standing Hypothesis 4.3 holds, and Steps 1 to 7 apply. If k is even, G is bipartite by Lemma 12.1. If k is odd, G is bipartite by Lemma 13.4. □

Proof of Corollary 2.7. If G has such an orientation, it is a k-certificate, and G is bipartite by Theorem 2.6. If G is bipartite, its maximum degree is k, and Theorem 2.5 gives a k-certificate. □

Remark 14.1 (what the proof used). Apart from elementary facts about walks and parity, the proof used three things: the star lemma (Lemma 5.1), the count that forces tightness (Lemma 6.1), and the fact that a directed odd cycle has no kernel (Lemma 3.23). It did not use that G is of class one, and it did not use the harder "sufficiency" half of the Borodin, Kostochka and Woodall characterisation [3], nor the stable roommates theory of Tan [11].

15 Worked examples

The examples follow the proof on small graphs. Every number here is printed by the script

`lecc-small/reverse_galvin_checks.py`.

Example 15.1 (the 5-cycle, $k = 2$)**.** Let $G = C_5$ with vertices $1, 2, 3, 4, 5$ in cyclic order. Each vertex has two edges, ranked 0 and 1. Tightness says $r_u(v) + r_v(u) = 1$, so on every edge exactly one end gives rank 0. Since $k = 2$ is even, $M_0 = \emptyset$, and O has the arc $u \to v$ exactly when $r_u(v) = 0$. Each vertex has exactly one out-arc and one in-arc (Lemma 8.6), so O is a directed 5-cycle. Read in its direction, the 5-cycle is admissible, hence a ring, hence an odd ring, and Lemma 7.2 says that D restricted to the five edges is a directed 5-cycle in $L(C_5) \cong C_5$, which has no kernel. Directly: $L(C_5)$ is a 5-cycle, and its only orientations with every out-degree at most 1 are the two directed 5-cycles, neither of which has a kernel. The brute-force search confirms: 2 orientations, 0 certificates.

Example 15.2 (K_4, $k = 3$)**.** K_4 is 3-regular, class one and not bipartite. The script finds that exactly 6 systems of rankings on K_4 satisfy the conclusions of Steps 1 and 2 (each vertex ranks

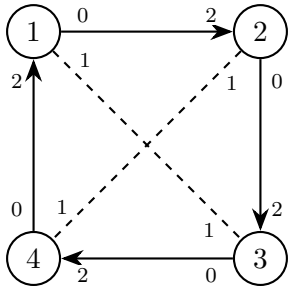


Figure 1: A tight rank system on K_4. The small number next to vertex w on an edge wv is $r_w(v)$. Solid arrows are the arcs of O; dashed edges form M_0. Every vertex ranks its clockwise successor 0, the vertex opposite 1, and its predecessor 2.

its neighbours $0, 1, 2$, and $r_u(v) + r_v(u) = 2$ on every edge), and that every one of them contains an odd ring. Figure 1 shows one of them:

w	$r_w(\cdot) = 0$	$r_w(\cdot) = 1$	$r_w(\cdot) = 2$
1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

Tightness holds: for example $r_1(2) + r_2(1) = 0 + 2 = 2$ and $r_1(3) + r_3(1) = 1 + 1 = 2$.

- *Step 4.* The middle rank is $\frac{k-1}{2} = 1$, so $M_0 = \{13, 24\}$, the two diagonals. Each of the other four edges uv has ranks 0 and 2 at its ends, and is oriented $u \to v$ when $r_u(v) = 0$. So O has the arcs $1 \to 2 \to 3 \to 4 \to 1$. Each vertex has one out-arc and one in-arc.

- *Step 7.* $G - M_0$ is the 4-cycle $1, 2, 3, 4$, which is bipartite, with col $= (0, 1, 0, 1)$ on $(1, 2, 3, 4)$.

- *Step 9.* Both diagonals join vertices of equal colour, so $\lambda(13) = \lambda(24) = 1$. There is one component H. The jump $(3, 1)$ is a jump cycle of length $t = 1$ with label sum 1. Here $v_1 = 3$ and $u_1 = 1$. The directed path of O from $u_1 = 1$ to $v_1 = 3$ is $P_1 = 1 \to 2 \to 3$, of length $2 \equiv \text{col}(1) + \text{col}(3) = 0$. Then $Z = P_1$ followed by $f_1 = 31$ is the triangle $1, 2, 3$, of length $3 \equiv \lambda(13) = 1$.

- *Step 5.* Z is admissible, so it should be a ring. Check: vertex 1 prefers 2 (rank 0) to 3 (rank 1); vertex 2 prefers 3 (rank 0) to 1 (rank 2); vertex 3 prefers 1 (rank 1) to 2 (rank 2). It is an odd ring.

- *Step 3.* In D, by Lemma 5.3(b): at 1, the edge 13 (rank 1) points to 12 (rank 0); at 2, 12 (rank 2) points to 23 (rank 0); at 3, 23 (rank 2) points to 31 (rank 1). So $12 \to 23 \to 31 \to 12$ is a directed triangle in D, with no kernel.

Directly, without any lemma: among the 38 orientations of $L(K_4)$ with every out-degree at most 2, none is kernel-perfect (Table 2).

Example 15.3 ($K_{3,3}$, where the certificates exist)**.** $K_{3,3}$ is 3-regular and bipartite, so Galvin's theorem gives 3-certificates. A brute-force search over all 148 orientations of $L(K_{3,3})$ with out-degree at most 2 finds exactly 12 kernel-perfect ones. There are exactly 12 proper edge colourings of $K_{3,3}$ with colours $0, 1, 2$ (they are the 3×3 Latin squares). This is what Theorem 16.3 predicts.

Example 15.4 (the diamond: why regularity is needed)**.** The diamond is K_4 minus one edge. It is not bipartite (it contains a triangle), not regular (degrees $3, 3, 2, 2$), class one with $\Delta = 3$, and of the form "bipartite plus an edge" (a 4-cycle plus a chord). A brute-force search finds

exactly 2 kernel-perfect orientations of its line graph with every out-degree at most 2, so it has 3-certificates. This agrees with McDonald's Theorem 3 [8], and shows that Theorem 2.6 fails without regularity. In terms of the proof, Step 2 fails: the slack of Remark 6.3 is 2, not 0.

16 Rigidity: on regular bipartite graphs, all certificates are Galvin's

Definition 16.1 (Galvin orientation)**.** Let G be a simple bipartite graph with bipartition (A, B) (every edge has one end in A and one in B), and let $c : E(G) \to \{0, 1, \ldots, k-1\}$ be a proper edge colouring. The *Galvin orientation* $D_{c,A}$ of $L(G)$ is defined as follows. Let e, f be distinct edges with common end w (unique by Lemma 4.2(b)); note $c(e) \neq c(f)$.

- If $w \in A$: $e \to f$ iff $c(e) > c(f)$ (towards the smaller colour).
- If $w \in B$: $e \to f$ iff $c(e) < c(f)$ (towards the larger colour).

This is the orientation of Galvin [5] and of [1], and it is McDonald's "Galvin orientation" with respect to (U, D, c) for $D = A$ and $U = B$ [8, Section 2]. (The choice of which side reads colours upwards is a convention; swapping A and B swaps it.)

Lemma 16.2. *If G is k-regular and bipartite, then in $D_{c,A}$ every edge has out-degree exactly $k-1$.*

Proof. Let $e = ab$ with $a \in A$, $b \in B$. At each vertex the k edges have k different colours from $\{0, \ldots, k-1\}$, so every colour occurs exactly once. At a, e points to the edges of smaller colour: there are $c(e)$ of them. At b, e points to the edges of larger colour: there are $k-1-c(e)$ of them. By the splitting argument of Lemma 5.3(c), which only uses simplicity, $d^+(e) = c(e) + (k-1-c(e)) = k-1$. □

Galvin's theorem, in the form we cite, says $D_{c,A}$ is kernel-perfect. So by Lemma 16.2, every Galvin orientation of a k-regular bipartite graph is a k-certificate. The rigidity theorem is the converse.

Theorem 16.3 (rigidity)**.** *Let G be a simple k-regular bipartite graph, $k \geq 1$, with bipartition (A, B).*

(a) *For every k-certificate D of G there is exactly one proper edge colouring $c : E(G) \to \{0, \ldots, k-1\}$ with $D = D_{c,A}$, namely $c(ab) = r_a(b)$ for $a \in A$, $b \in B$.*

(b) *Consequently, using Galvin's theorem, $c \mapsto D_{c,A}$ is a bijection from the proper edge colourings of G with colours $0, \ldots, k-1$ onto the k-certificates of G.*

Proof. (a) Let D be a k-certificate. Standing Hypothesis 4.3 holds, so Steps 1 and 2 apply: ranks exist and are tight. Define $c(ab) = r_a(b)$ for every edge ab with $a \in A$, $b \in B$.

(1) *c is proper at vertices of A.* On $E(a)$, c equals r_a, which is a bijection onto $\{0, \ldots, k-1\}$ (Lemma 5.3(a)). So the edges at a have distinct colours.

(2) *c is proper at vertices of B.* By tightness, $c(ab) = r_a(b) = k-1-r_b(a)$. On $E(b)$, r_b is a bijection onto $\{0, \ldots, k-1\}$, and so is $x \mapsto k-1-x$; so c is injective on $E(b)$.

(3) *$D = D_{c,A}$ at vertices of A.* For distinct $e, f \in E(a)$: $e \to f$ in D iff $r_a(e) > r_a(f)$ (Lemma 5.3(b)) iff $c(e) > c(f)$.

(4) *$D = D_{c,A}$ at vertices of B.* For distinct $e, f \in E(b)$: $e \to f$ in D iff $r_b(e) > r_b(f)$ iff $k-1-c(e) > k-1-c(f)$ iff $c(e) < c(f)$.

(5) Every arc of $L(G)$ lies at exactly one common end (Lemma 4.2(b)), which is in A or in B. By (3) and (4), D and $D_{c,A}$ agree on every arc.

(6) *Uniqueness.* Suppose $D = D_{c',A}$ for a proper colouring c'. Let $a \in A$ and $e \in E(a)$. In $D_{c',A}$, e points to exactly the edges of $E(a)$ of smaller colour. The colours on $E(a)$ are $0, \dots, k-1$, each once, so the number of smaller colours is $c'(e)$. Hence $r_a(e) = c'(e)$, that is $c' = c$ on $E(a)$. Every edge has an end in A, so $c' = c$.

(b) By Galvin's theorem and Lemma 16.2, $c \mapsto D_{c,A}$ maps colourings to certificates. By the uniqueness in (a) it is injective (if $D_{c,A} = D_{c',A}$ then $c = c'$), and by the existence in (a) it is onto. □

Remark 16.4. Part (a) does not use Galvin's theorem: it says that *if* a certificate exists, it has Galvin's form. Only part (b), the statement that every Galvin orientation really is kernel-perfect, uses Galvin's theorem. The brute-force table (Table 2) checks the count in (b) independently of both: for C_4, C_6, $K_{3,3}$ and the cube Q_3 the number of certificates equals the number of colourings (2, 2, 12, 24).

17 Orientations with arcs in both directions

Some authors (for instance McDonald and Puleo [9]) apply the kernel lemma to digraphs in which two adjacent vertices may be joined by arcs in both directions. The kernel lemma remains true in that setting, since its standard proof never uses that at most one arc joins two vertices. The main theorem extends to it.

Proposition 17.1. *Let G be a simple k-regular graph, $k \geq 1$. Let D be a digraph with vertex set $E(G)$ such that (i) D has an arc between e and f only if e, f are adjacent in $L(G)$; (ii) every two adjacent e, f are joined by at least one arc, possibly by both; (iii) D is kernel-perfect; (iv) $d^+_D(e) \leq k-1$ for every e. Then no pair is joined by both arcs, so D is an orientation of $L(G)$ and a k-certificate. In particular G is bipartite.*

Proof. **(1)** By (i) and (ii), $|A(D)| = |E(L(G))| + p$, where p is the number of adjacent pairs joined by both arcs.

(2) By Lemma 3.14 and (iv), $|A(D)| = \sum_e d^+_D(e) \leq |E(G)|(k-1)$.

(3) By steps (2) and (3) of the proof of Lemma 6.1, $|E(L(G))| = |E(G)|(k-1)$.

(4) Hence $|E(G)|(k-1) + p \leq |E(G)|(k-1)$, so $p = 0$.
Then D is an orientation of $L(G)$ satisfying Definition 2.4, and Theorem 2.6 applies. □

18 Impact and consequences

This section explains what the theorem implies, what it connects to, and what it leads to. Results of other authors are cited, not proved.

18.1 The kernel method on regular graphs is exactly Galvin's theorem

Corollary 2.7 and Theorem 16.3 together give a complete description of the kernel method with lists of size Δ on simple regular graphs:

- it succeeds exactly on the bipartite ones (Corollary 2.7);
- where it succeeds, the certificates are exactly Galvin's orientations, one for each proper Δ-edge colouring (Theorem 16.3).

So for regular graphs, Galvin's 1995 construction is not one successful instance of the kernel method among possibly many others. It *is* the kernel method, completely. Any proof of the List Edge Colouring Conjecture for a non-bipartite regular graph must use a different idea, or the kernel lemma with lists larger than Δ (which cannot give the exact value $\chi' = \Delta$ for class one graphs).

18.2 Which cases of the conjecture are affected

For a simple regular graph G there are three cases.

(1) *G bipartite.* Then G is class one by König's theorem [7], the conjecture holds by Theorem 2.5, and the kernel method is the proof.

(2) *G of class two.* Then $\mathrm{ch}'(G) \geq \chi'(G) = \Delta + 1$, so no proof of $\mathrm{ch}'(G) = \Delta$ exists by any method, and Theorem 2.6 says nothing new. (Such G is never bipartite, by (1).)

(3) *G non-bipartite and of class one.* The conjecture predicts $\mathrm{ch}'(G) = \Delta$. **This is the case Theorem 2.6 is about**: the kernel method cannot prove the prediction, for any such G.

Case (3) is large. It contains, among others:

- *Complete graphs of even order.* K_{2n} for $n \geq 2$ is $(2n-1)$-regular, class one and not bipartite. The conjecture for K_{2n} is open in general; it is known when $2n-1$ is an odd prime (Schauz [10], via the Alon and Tarsi method) and for some further families (Jafari [6]). Theorem 2.6 recovers our earlier theorem that $L(K_{2n})$ has no $(2n-1)$-certificate, and it strengthens McDonald's Theorem 5 [8] for even order: McDonald excludes certificates of Galvin's form, Theorem 2.6 excludes all certificates. (For odd order, K_{2n+1} is class two, the relevant budget is $\chi' = \Delta + 1$, and Theorem 2.6 does not apply.)

- *Planar regular class one graphs that are not bipartite*, such as K_4, the triangular prism and the pentagonal prism $C_5 \square K_2$. Ellingham and Goddyn [4] proved the conjecture for every d-regular d-edge-colourable planar graph, using the Alon and Tarsi method. By Theorem 2.6, the kernel method could not have given this theorem for the non-bipartite ones.

- *Schauz's open small graphs.* Schauz [10] settled the conjecture for most graphs on at most 10 vertices with the Alon and Tarsi method, leaving open the regular class one graphs on which his method's relevant coefficient is zero (3 on 8 vertices and 51 on 10; we reproduced this census). Of these, 52 are covered by no known theorem; in particular none of the 52 is bipartite, so Theorem 2.6 shows the kernel method fails on every one. The Alon and Tarsi method fails on them too, for a reason of the same counting kind as Step 2: for k-regular G the graph polynomial of $L(G)$ has degree $|E(L(G))| = |E(G)|(k-1)$ in $|E(G)|$ variables, so the only monomial with every exponent at most $k-1$ is the one with every exponent equal to $k-1$, and on these graphs its coefficient is zero. So both standard methods provably fail on all 52. This explains why they are open, and it is the reason a third method was needed: the SAT-based search recorded in `lecc-small/RESULTS.md`, which settles 6 of them.

- *Cubic class one graphs containing a triangle.* Our earlier triangle theorem says these have no 3-certificate. It is the special case of Theorem 2.6 in which the odd cycle has length 3 and $k = 3$.

18.3 Separating the two standard methods

The two standard tools for the conjecture are the kernel method and the Alon and Tarsi polynomial method [2]. Theorem 2.6 shows they are incomparable on regular graphs.

- The Alon and Tarsi method proves $\mathrm{ch}'(K_4) = 3$, and more generally the Ellingham and Goddyn theorem; by Theorem 2.6 the kernel method proves neither for non-bipartite graphs.
- Conversely, for $K_{3,3}$ the kernel method works (Galvin), while the Alon and Tarsi method with lists of size 3 does not: by the same counting, the only usable monomial is the one with every exponent 2, and its coefficient is zero. (Up to sign it is the difference between the numbers of even and odd Latin squares of order 3, which are equal for every odd order.) Our earlier computation confirms this.

Any attack on case (3) above therefore has to start from the Alon and Tarsi side, or from a new method.

18.4 Why regularity is the right hypothesis

Remark 6.3 shows that regularity is used in exactly one place, to make the budget tight, and Example 15.4 shows that the theorem is false without it. A small consistency check with McDonald's Theorem 3: if a bipartite graph plus one edge xy is k-regular and not bipartite, then in the bipartite graph x and y are on the same side, X say, with degree $k-1$ while all other vertices have degree k; counting edges from each side gives $k|X| - 2 = k|Y|$, so k divides 2. For $k = 1$ adding xy keeps the graph bipartite, so $k = 2$, and the graph is a union of cycles with one odd cycle, which is class two. So McDonald's positive result never produces a regular non-bipartite class one graph, exactly as Theorem 2.6 requires.

18.5 The limits of the statement

- *It is not a lower bound on* ch'. It does not say $\mathrm{ch}'(G) > \Delta$ for non-bipartite regular G; K_4 is a counterexample to that stronger statement (Section 2.4).
- *Simple graphs only.* The multigraph case is open; see Remark 5.4 for exactly where the proof uses simplicity.
- *Regular graphs only.* The diamond (Example 15.4) shows this is necessary.
- *Budget* Δ *only.* With lists of size $\Delta + 1$ the kernel method succeeds on many non-bipartite graphs, for example the Petersen graph and all graphs without odd cycles of length at least 5 [8, Theorems 4 and 8]. For class one graphs this does not give the conjectured value.
- *The kernel lemma only.* The statement is about Lemma 2.3. It says nothing about other methods that use orientations, such as the Alon and Tarsi method.

18.6 Questions it leads to

Question 18.1 (multigraphs)**.** *Is Theorem 2.6 true for regular multigraphs? Galvin's theorem holds for bipartite multigraphs, so the natural statement is again a characterisation. The proof breaks at Lemma 5.3(c), where parallel edges share two stars.*

Question 18.2 (smallest budget)**.** *For a non-bipartite regular graph G of class one, what is the least k for which G has a k-certificate? Theorem 2.6 says it is at least $\Delta + 1$. For K_4 it is exactly 4, since McDonald exhibits a Galvin orientation with respect to 4 [8]. This is the regular-graph case of McDonald's question for Galvin orientations, now asked for all orientations.*

Question 18.3 (near-regular graphs)**.** *Remark 6.3 measures the failure of regularity by the slack $\sum_{uv}\big(\Delta - \frac{d(u)+d(v)}{2}\big)$. When the slack is small, the budget is almost tight, and Steps 3 to 9 may survive on most of the graph. Is there a quantitative version of Theorem 2.6, for example*

one saying that every odd cycle far enough from the vertices of degree less than Δ forbids a Δ-certificate?

Question 18.4 (a method for case (3))**.** *Both standard methods fail on Schauz's 52 open graphs. Is there a structural reason, beyond these small cases, that makes non-bipartite regular class one graphs hard, or a third general method that works on them?*

18.7 Novelty

We have not found Theorem 2.6 or Theorem 16.3 in the literature. We checked the characterisation of kernel-perfect line graph orientations by Borodin, Kostochka and Woodall [3] (it characterises the orientations, not which graphs admit one within budget Δ), McDonald [8] (negative results only for Galvin orientations, for K_n and for one non-regular graph), McDonald and Puleo [9] (consistent with Proposition 17.1; no non-bipartite regular graph with $k \geq 3$ lies in their class), the papers citing [3], and recent work on the conjecture for complete graphs [6]. Two sources could not be checked: the 2016 Auburn M.Sc. thesis of A. Blumenthal, *On Galvin's Theorem and Stable Matchings*, whose server was unreachable on 9 and 12 October 2026, and Woodall's 2001 survey of list colourings. Since the proof is short and elementary, the result may be known to specialists without being published. It should be described as "not found in the literature", not as new.

19 Computational confirmation

The proof does not depend on any computation. The computations below check it independently.

Brute force, using no lemma. The script `lecc-small/reverse_galvin_checks.py` enumerates every orientation of $L(G)$ with every out-degree at most $k-1$, and tests kernel-perfectness directly by searching every induced subdigraph for a kernel. It uses none of the lemmas of this document. It also counts the proper edge colourings with colours $0, \ldots, k-1$ by backtracking. The output is Table 2.

G	k	bip.	orient., $d^+ \leq k-1$	certificates	colourings
C_4	2	yes	2	2	2
C_5	2	no	2	0	0
C_6	2	yes	2	2	2
C_7	2	no	2	0	0
K_4	3	no	38	0	6
$K_{3,3}$	3	yes	148	12	12
triangular prism $K_3 \square K_2$	3	no	154	0	6
cube Q_3	3	yes	600	24	24
pentagonal prism $C_5 \square K_2$	3	no	2466	0	30
Möbius ladder on 8 vertices	3	no	594	0	18
Controls outside the hypotheses of Theorem 2.6:					
diamond $K_4 - e$ (not regular)	3	no	72	2	6
C_3 with $k = \Delta + 1$	3	no	8	6	6
C_5 with $k = \Delta + 1$	3	no	32	30	30

Table 2: Brute-force counts. "Certificates" are the kernel-perfect orientations among the previous column. "Colourings" are the proper edge colourings with colours $0, \ldots, k-1$. Every regular non-bipartite graph has 0 certificates even when it is class one (a non-zero colouring count), as Theorem 2.6 says. Every regular bipartite graph has as many certificates as colourings, as Theorem 16.3 says.

The K_4 rank systems. The same script lists all 6 tight rank systems on K_4 and confirms that each contains an odd ring (Example 15.2).

SAT search on 1923 graphs. The script `lecc-small/kernel_sat.py` decides, with a SAT solver, whether a graph has a k-certificate. It encodes the star lemma and the odd-ring characterisation (the latter is the known theorem of Borodin, Kostochka and Woodall [3], cross-checked against brute force in `lecc-small/tests/`). It was run on every connected regular graph on 6 and 8 vertices with degree at least 3, on 10 vertices with degree from 3 to 7, on all cubic graphs on 12 vertices, all triangle-free cubic graphs on 14 vertices, and all 4-regular graphs on 12 vertices: 1923 graphs in total. A certificate exists for exactly the 29 bipartite ones, and for none of the 1894 non-bipartite ones (`data/kernel_all_small.csv`, `data/kernel_N12_k4.csv`).

A Verification checklist

Each row is one claim of the proof of Theorem 2.6 or Theorem 16.3, with the results it depends on. A verifier can tick the rows in order.

Result	Content	Depends on
Lemma 3.1	$0 + \cdots + (k-1) = k(k-1)/2$	arithmetic
Lemma 3.2	a sum at its maximum has every term at the maximum	arithmetic
Lemma 3.4	two edges share at most one end	Definition 3.3
Lemma 3.5	$\sum d(v) = 2\lvert E\rvert$	double counting
Lemma 3.7	odd closed walk contains odd cycle	induction
Lemma 3.9	length parity equals colour sum	induction
Lemma 3.11	bipartite iff no odd cycle	3.7, 3.9
Lemma 3.12	bipartite iff components are	Definition 3.10
Lemma 3.14	$\sum d^+ = \lvert A\rvert = \sum d^-$	double counting
Lemma 3.17	directed version of 3.7	induction
Lemma 3.18	shortest directed walks are paths	minimality
Lemma 3.20	balanced: every arc has a return path	3.18, counting
Lemma 3.21	balanced: strongly connected inside components	3.20, 3.18
Lemma 3.22	kernels are non-empty	Definition 2.2
Lemma 3.23	directed odd cycle has no kernel	Definition 2.2
Lemma 3.24	tournament kernels are sinks	3.22
Lemma 3.25	kernel-perfect tournaments are transitive	3.23
Lemma 3.26	ranks in transitive tournaments	3.25
Lemma 4.2	stars are cliques; edge count of $L(G)$	3.4
Lemma 5.1 (Step 1)	star lemma	4.2, 3.25
Lemma 5.3 (Step 1)	ranks; $d^+(uv) = r_u(v) + r_v(u)$	5.1, 3.26, 3.4
Lemma 6.1 (Step 2)	$r_u(v) + r_v(u) = k - 1$	4.2, 3.5, 3.2, 5.3
Lemma 7.2 (Step 3)	no odd ring	5.3, 3.23
Lemma 8.2 (Step 4)	$y(v,u) = -y(u,v)$	6.1

Result	Content	Depends on
Lemma 8.4 (Step 4)	O orients $G - M_0$	8.2
Lemma 8.5 (Step 4)	M_0 empty or a perfect matching	5.3
Lemma 8.6 (Step 4)	O balanced	5.3, 8.2
Lemma 9.2 (Step 5)	admissible iff $y \leq 0$ along the cycle	Definition 8.3
Lemma 9.3 (Step 5)	admissible cycles are rings	8.2, 9.2, 8.5
Corollary 9.4 (Step 5)	no admissible odd cycle	9.3, 7.2
Corollary 9.5 (Step 5)	O has no odd directed closed walk	9.4, 3.17
Lemma 10.1 (Step 6)	directed paths inside components	8.6, 3.21, 3.20
Lemma 11.1 (Step 7)	components of $G - M_0$ bipartite	3.11, 10.1, 9.5
Lemma 12.1 (Step 8)	k even $\Rightarrow$ bipartite	8.5, 11.1, 3.12
Lemma 13.2 (Step 9)	even jump cycles $\Rightarrow$ bipartite	(13.1)
Lemma 13.3 (Step 9)	shortest odd jump cycle: distinct components, $u_i \neq v_i$	13.2, 8.5
Lemma 13.4 (Step 9)	k odd $\Rightarrow$ bipartite	13.3, 10.1, 3.9, 9.4
Theorem 2.6	main theorem	12.1, 13.4
Lemma 16.2	Galvin orientations have $d^+ = k - 1$	5.3(c) argument
Theorem 16.3(a)	certificates are Galvin orientations	5.3, 6.1, 4.2
Theorem 16.3(b)	bijection with colourings	(a), 16.2, Theorem 2.5 (cited)
Proposition 17.1	arcs in both directions excluded	3.14, 6.1, 2.6

B Notation

$E(w)$, $d(w)$	edges at w; degree of w
$L(G)$	line graph: vertices are edges of G, adjacent iff they share an end
$e \to f$	arc of the certificate D (Steps 1 to 3), or of O (Steps 4 to 9)
d^+, d^-	out-degree, in-degree
k-certificate	kernel-perfect orientation of $L(G)$ with every out-degree $\leq k - 1$
$r_w(v) = r_w(wv)$	rank of the edge wv at w; 0 is best
ring	cycle on which every vertex prefers its successor to its predecessor
$y(u, v)$	$r_u(v) - \frac{k-1}{2}$; $y(v, u) = -y(u, v)$
M_0	edges with $y = 0$ (middle rank at both ends)
O	orientation of $G - M_0$, $u \to v$ iff $y(u, v) < 0$
admissible	every edge in M_0 or traversed forwards in O
col, comp(x)	proper 2-colouring of $G - M_0$; component of $G - M_0$ containing x
$\lambda(f)$	1 if the ends of $f \in M_0$ have equal colour, else 0
jump, jump cycle	move along an edge of M_0; closed sequence of jumps
$D_{c,A}$	Galvin orientation from colouring c and side A

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